UNIVERSITY OF CAMBRIDGE

Machine Learning and the Physical World

Lecture 7 : Sequential Decision Making - Bayesian Optimisation

Carl Henrik Ek - che29@cam.ac.uk

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http://carlhenrik.com

Re-Cap

- The infeasibility of truth and the search for knowledge

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- Parametrise our ignorance/beliefs

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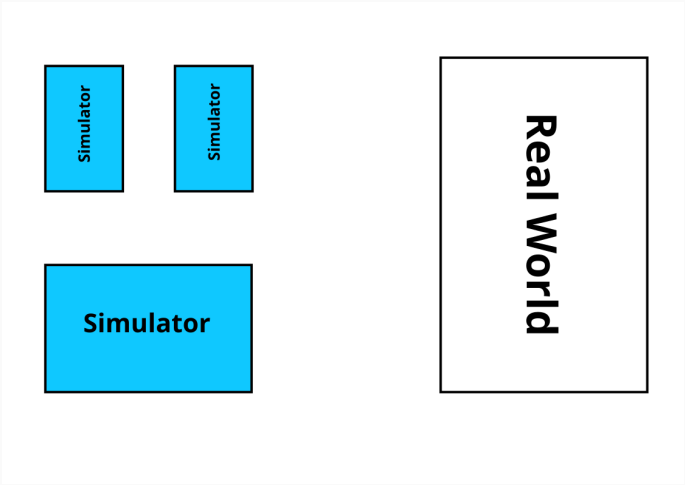
- Parametrise our ignorance/beliefs
- Statistical Inference to update our knowledge from experiment
- Simulation and Emulation

Simulation and Emulation



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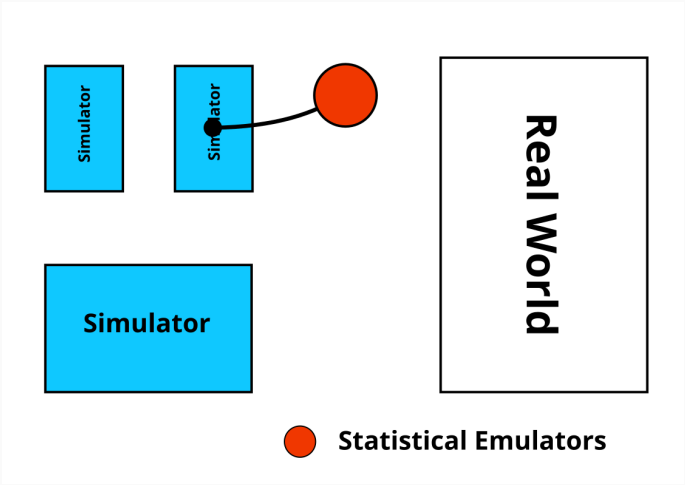

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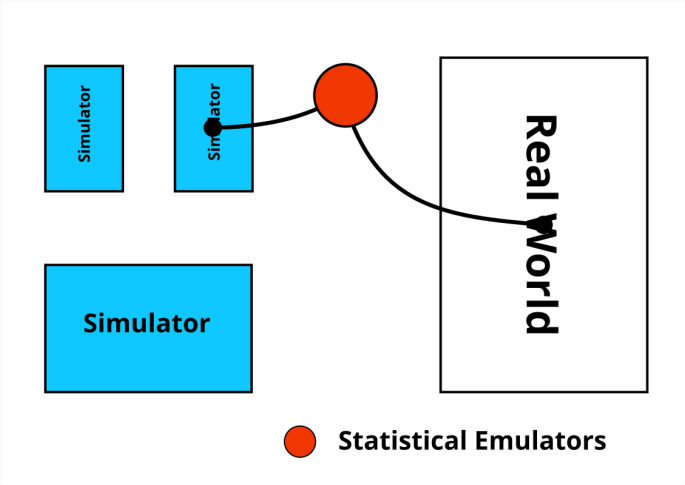

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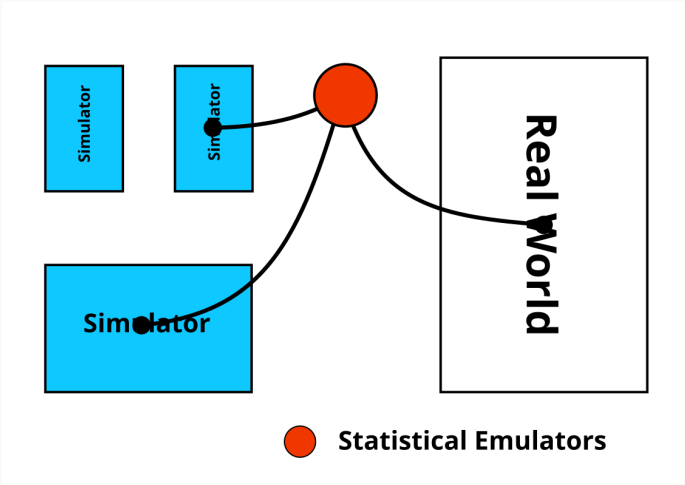
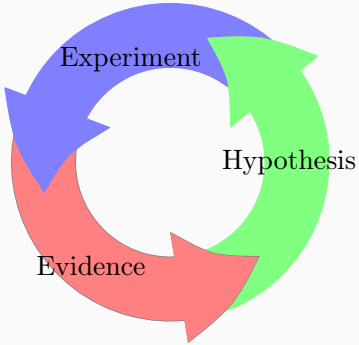


$$\hat{f} = \underset{f \in \mathcal{F}}{\operatorname{argmin}} \int \ell(\mathcal{A}_{\mathcal{F}}(\mathcal{S}), x, y) p(x, y) \mathrm{d}x \mathrm{d}y$$

- $\mathcal{F}$ space of functions
- $\mathcal{A}$ learning algorithm
- $\mathcal{S} \sim P(\mathcal{X} \times \mathcal{Y})$
- $\ell(\mathcal{A}_{\mathcal{F}}(\mathcal{S}), x, y)$ loss function



$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{\int p(y \mid \theta)p(\theta)\mathrm{d}\theta}$$

Active Learning

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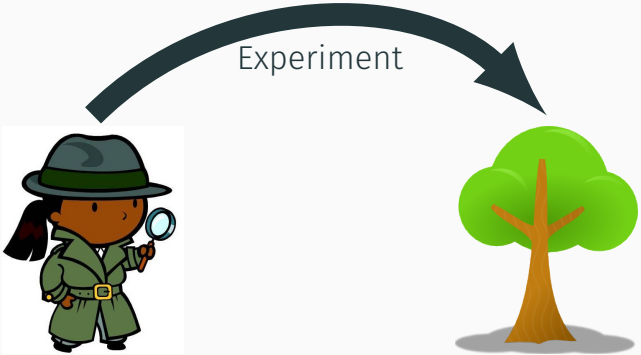

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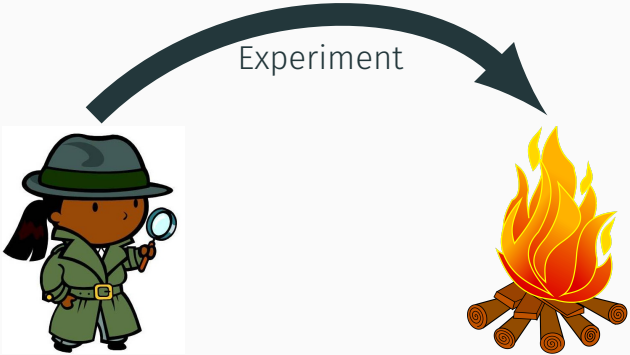

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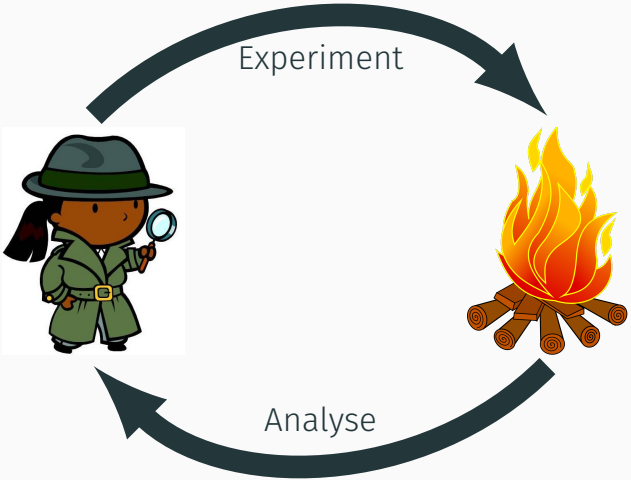

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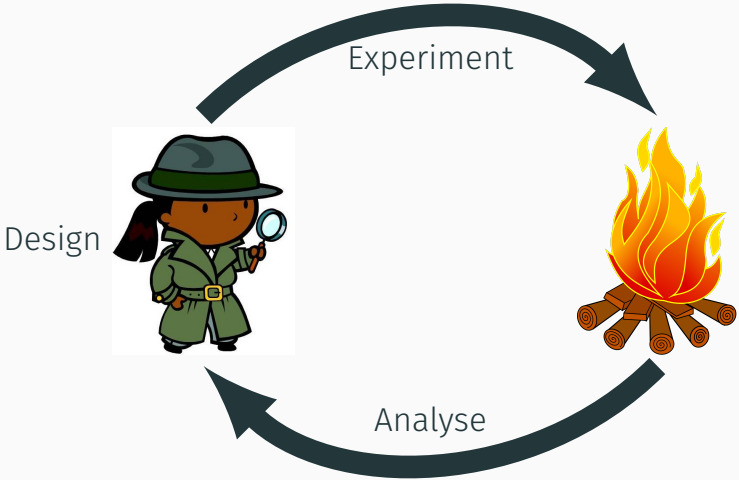
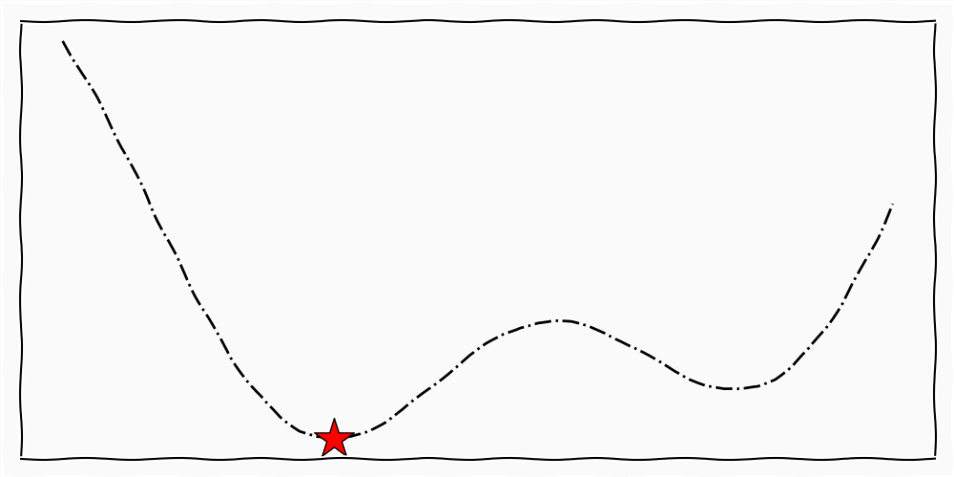


PUB QUIZ

Finding Extremum of a function



Today

Black-Box Optimisation how can we find the extremum of an explicitly unknown function?

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Surrogate Models how can we build a model as a surrogate for the unknown function?

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Black-Box Optimisation how can we find the extremum of an explicitly unknown function?

Surrogate Models how can we build a model as a surrogate for the unknown function?

Sequential decision making how can we come up with a strategy for sequentially exploring the function?

Bayesian Optimisation

Black-box Optimisation

$$x^{(*)} = \operatorname*{argmin}_{x \in \mathcal{X}} f(x)$$

- $\mathcal{X}$ is a bounded domain

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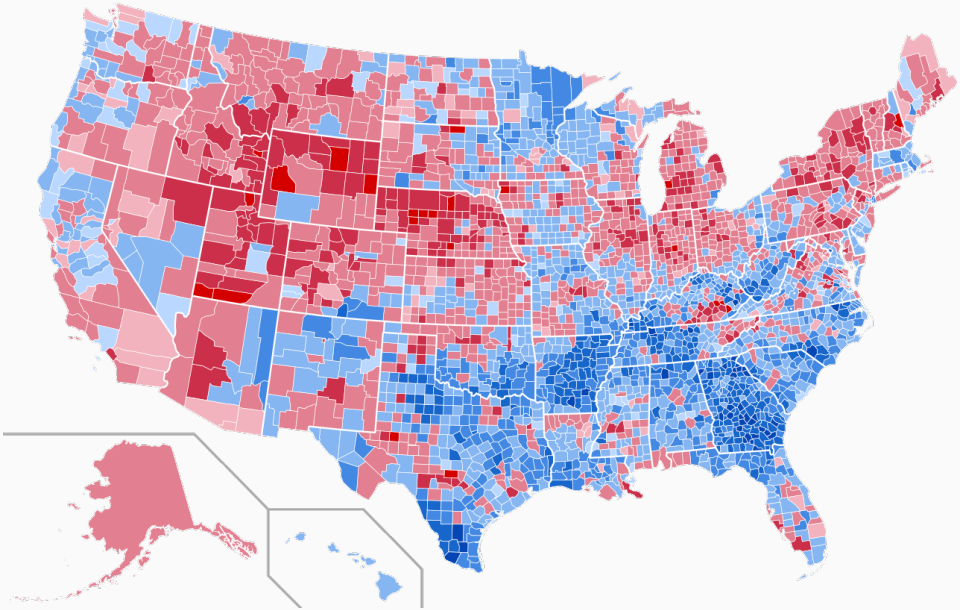
- f is explicitly unknown
- Evaluations of f may be noisy
- Evaluations of f is expensive



Random Search [Brochu et al., 2010]

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$$f(x^{(-)}) \leq f(x^{(*)}) - \epsilon$$

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$$\|f(x_1) - f(x_2)\| \leq C\|x_1 - x_2\|$$

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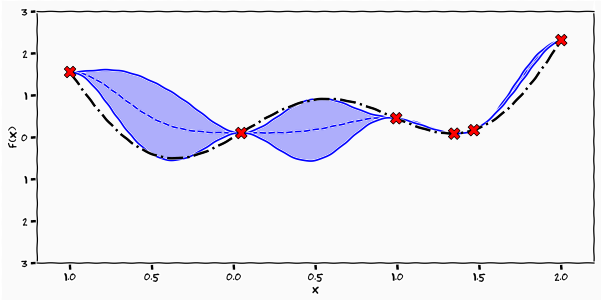
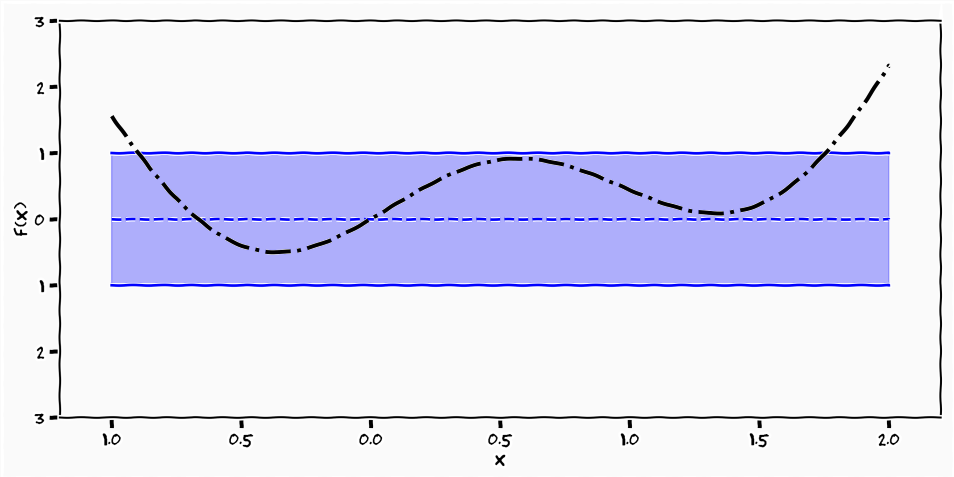
- Requires $\left(\frac{C}{2\epsilon}\right)^d$ evaluations on a d -dimensional hypercube
- Surrogate model $p(f)$

Gaussian Process Surrogate



- allows for principled priors and narrow priors
- provides belief over the whole domain

Posterior Search: Random



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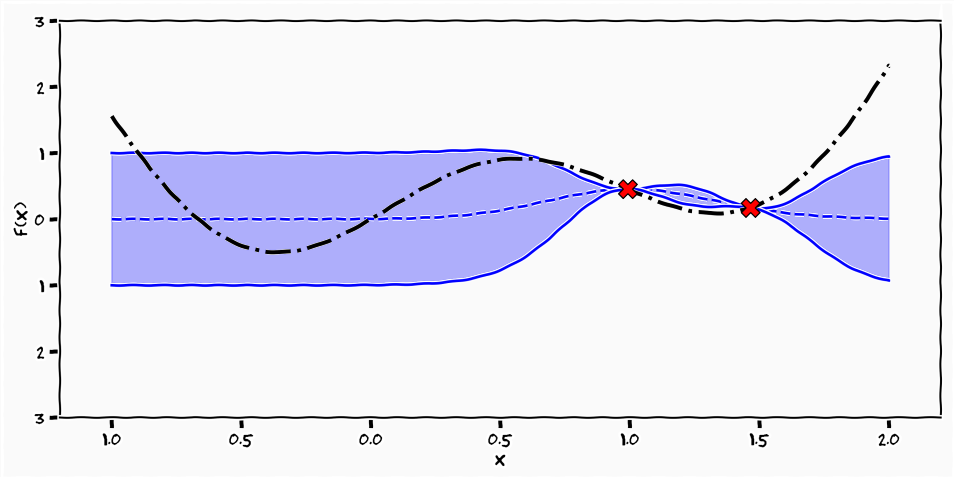

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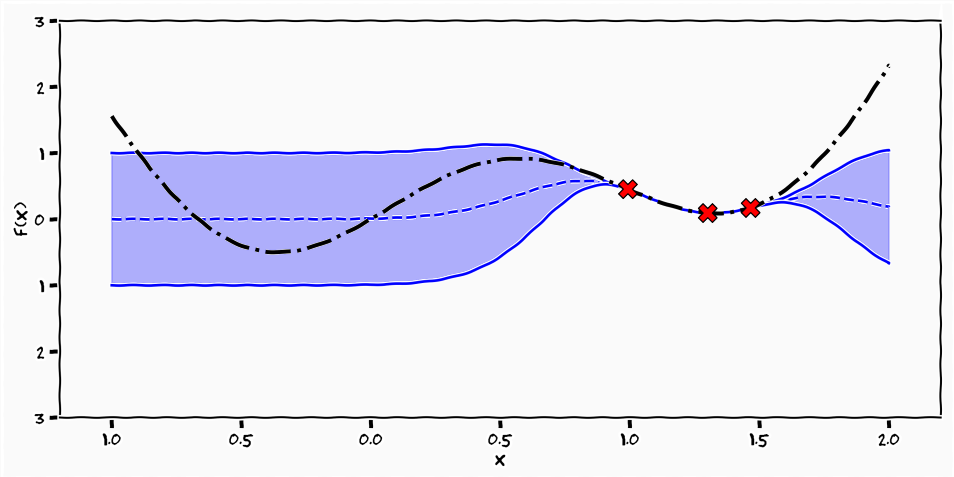

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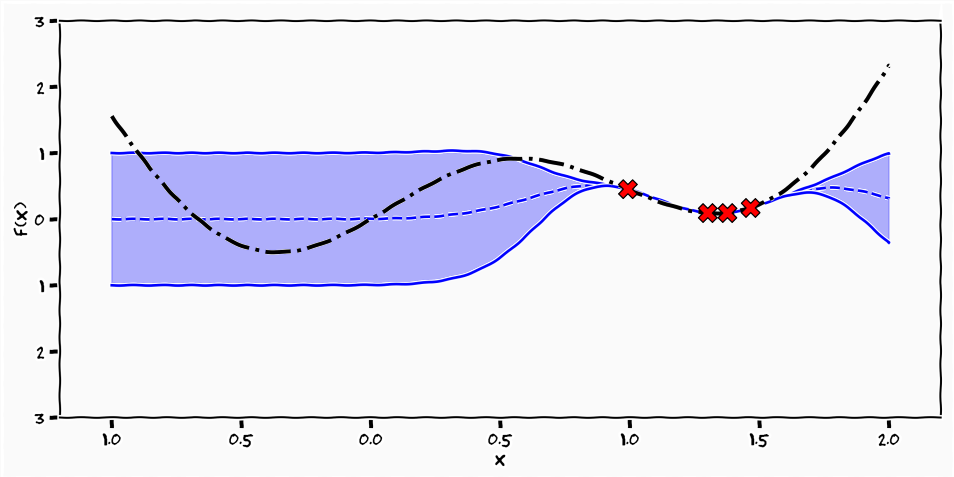

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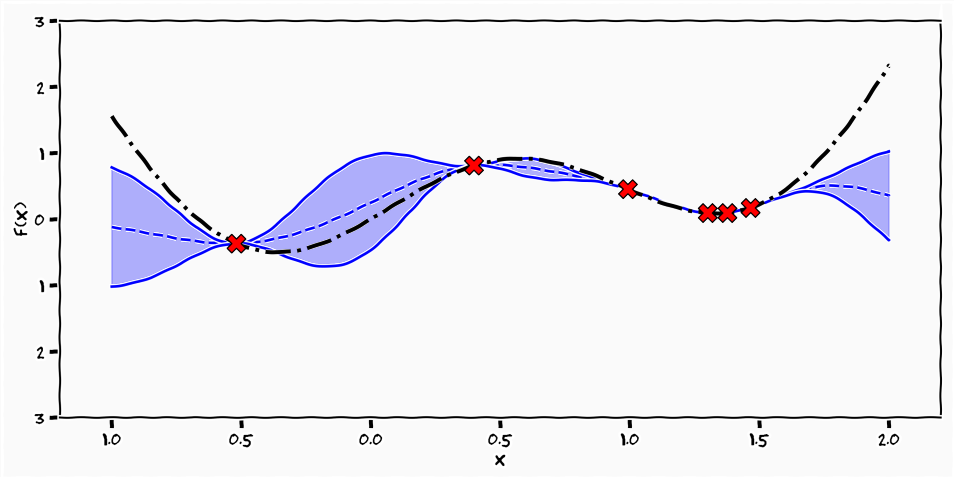

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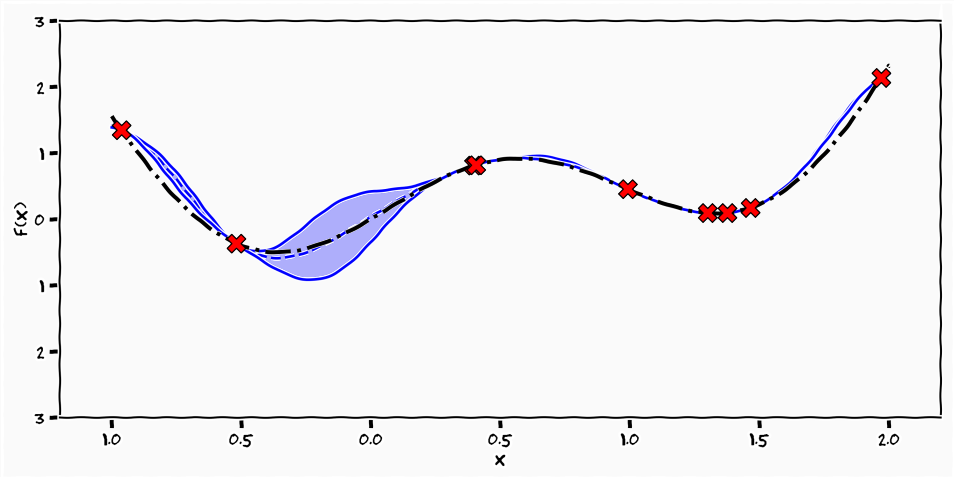
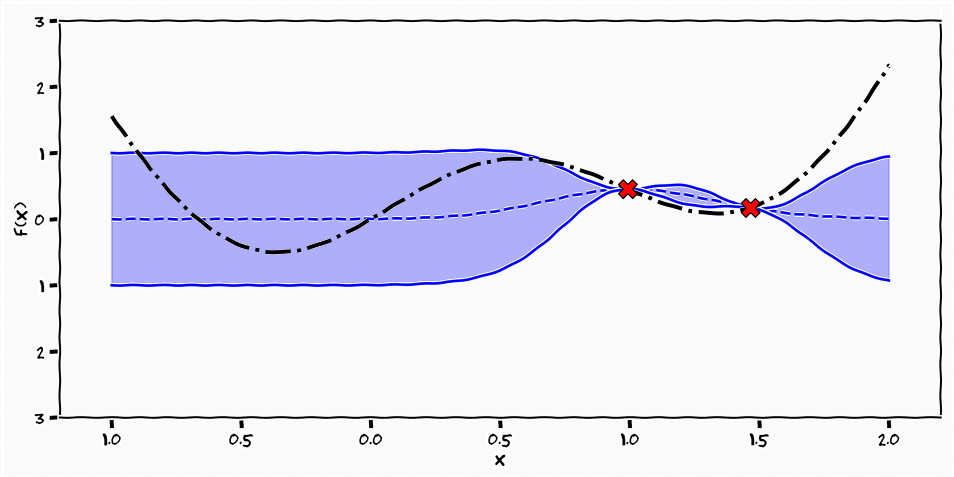


Posterior Search: Min



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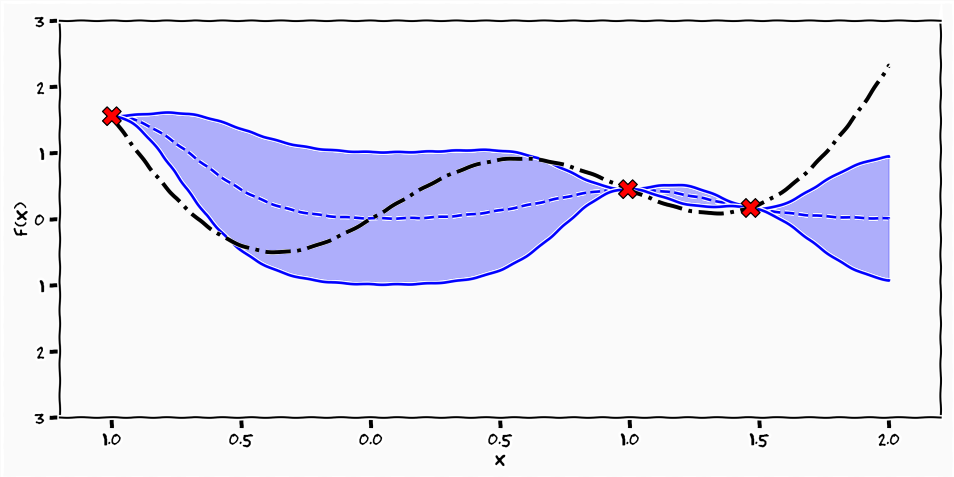

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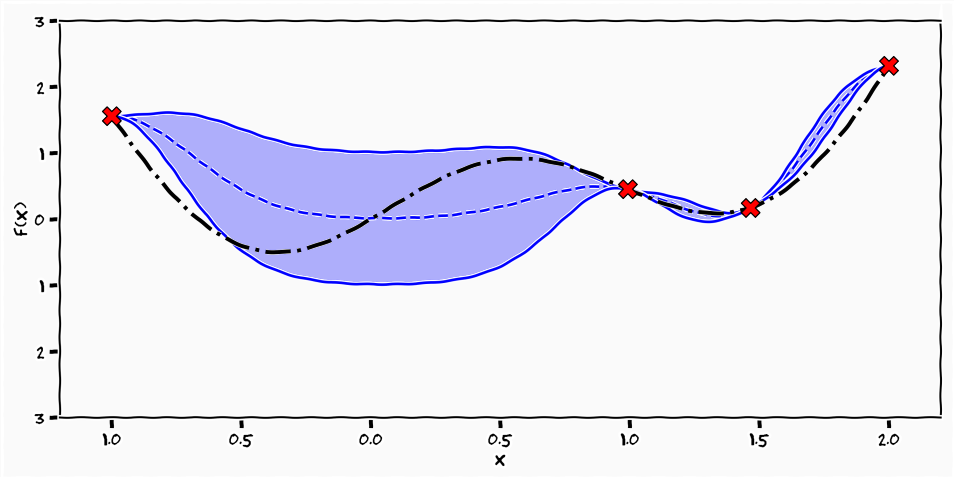

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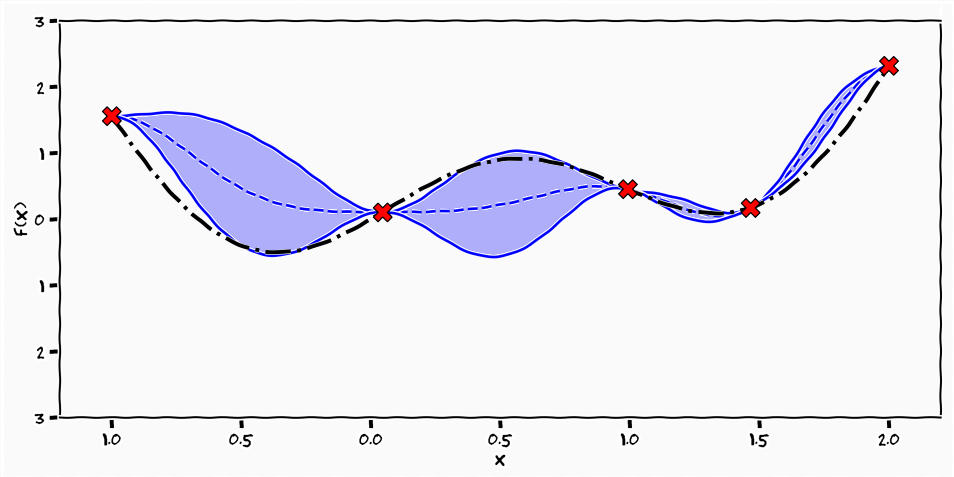

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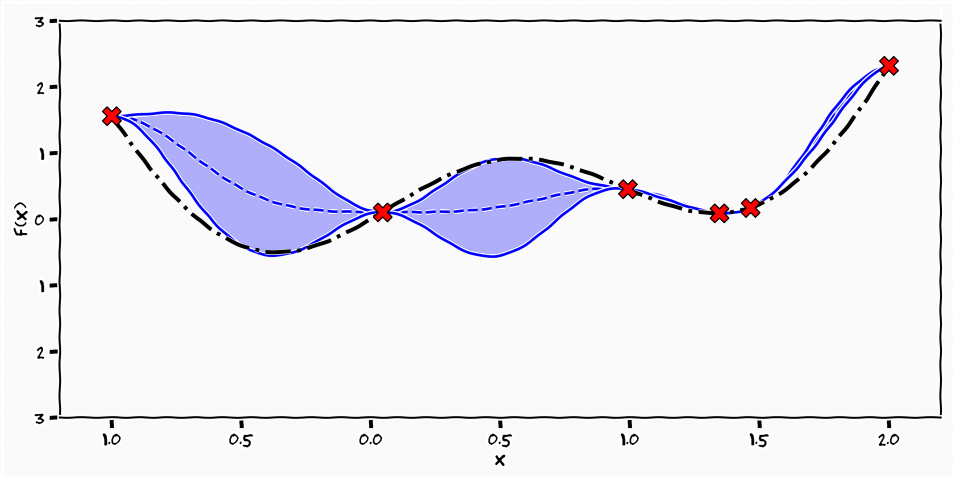

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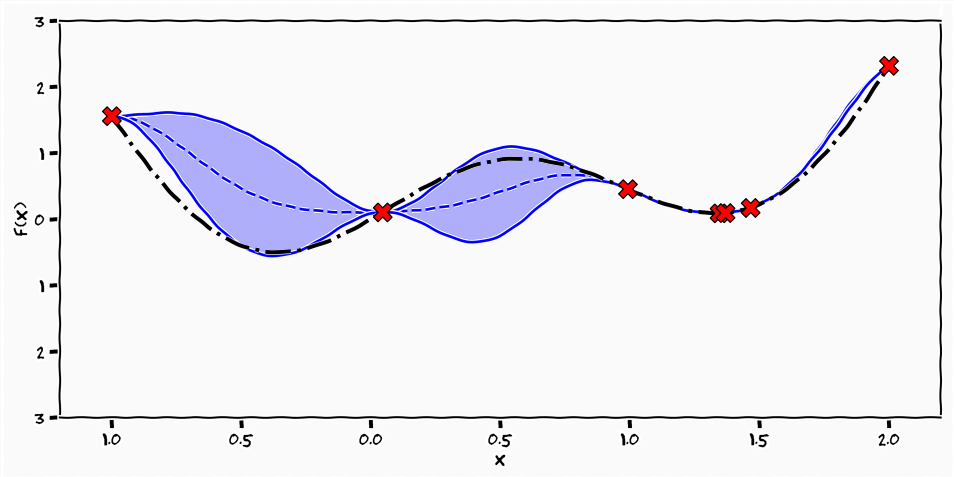

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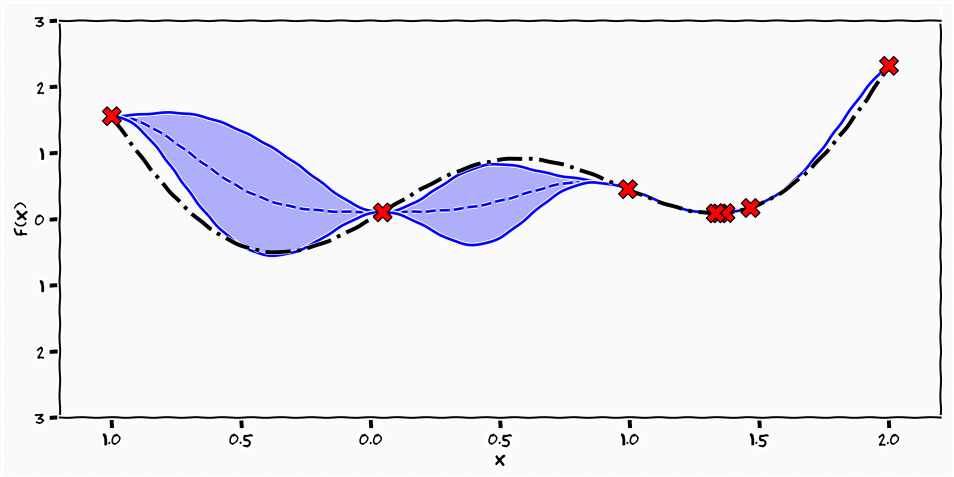

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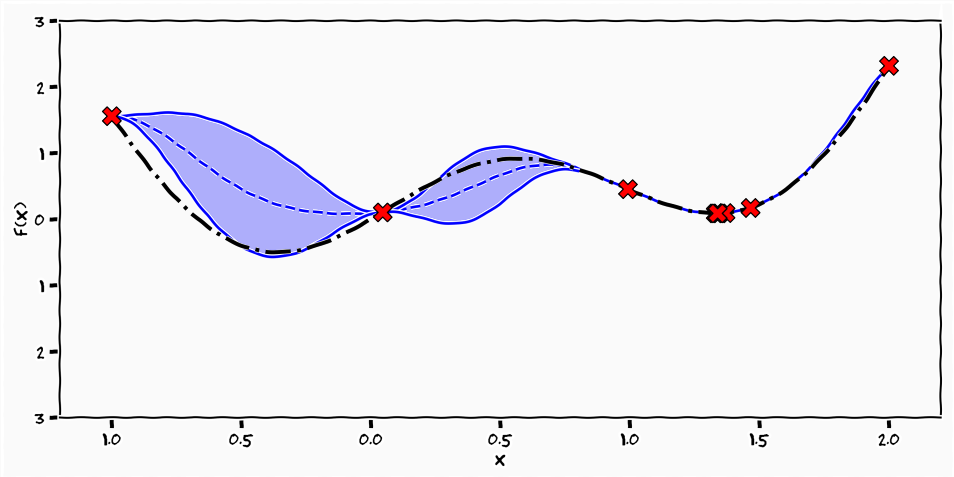


STRONGARM
CAMERONS
TIMOTHY TAYLOR'S
LANDLORD
FURNACE
NICK STAFFORD'S
West Coast
Mosaic IPA
4.0% ABV

Coca-Cola

Coca-Cola

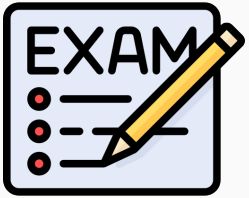
WYE VALLEY
BITTER
Superior Quality
WYE VALLEY
BREWERY
WYE VALLEY
BREWERY

Coca-Cola

WYE VALLEY
BITTER
Superior Quality
WYE VALLEY
BREWERY
WYE VALLEY
BREWERY

ALASKA
DISTILLERY
HANDCRAFTED
SMOKED SALMON
FLAVORED VODKA

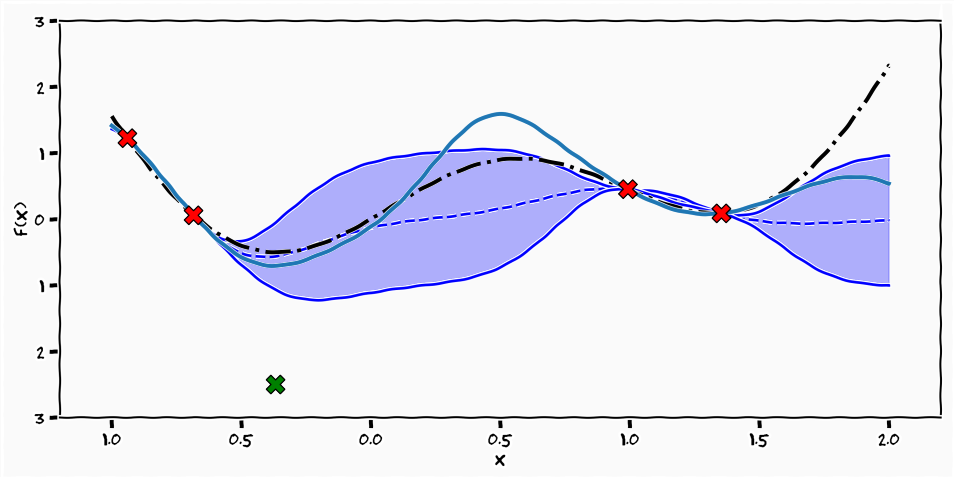
Exploration and Explotation



Exploitation use the knowledge that we currently have

Exploration try to gain new knowledge by trying new things

Surrogate Uncertainty



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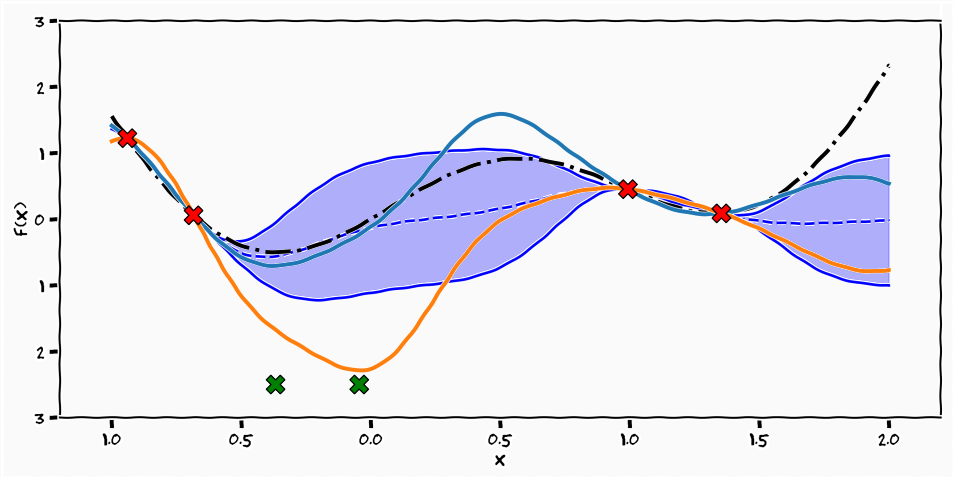

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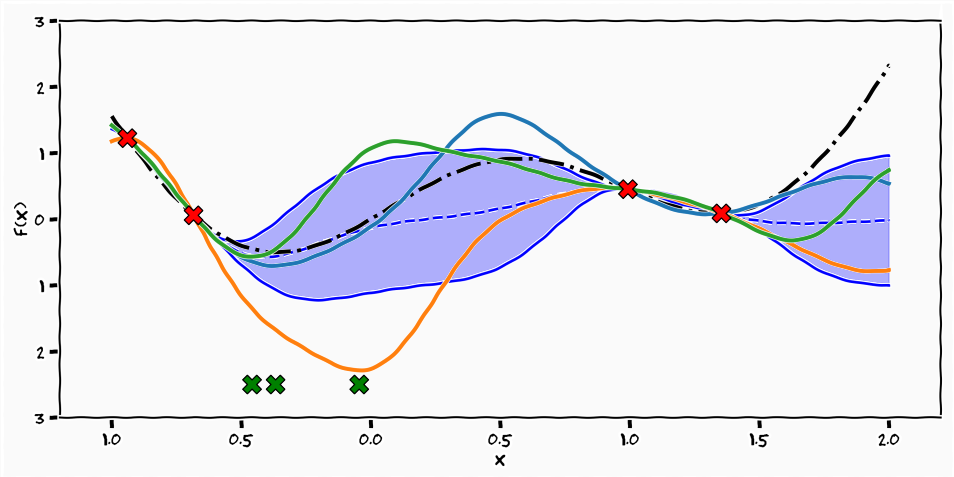

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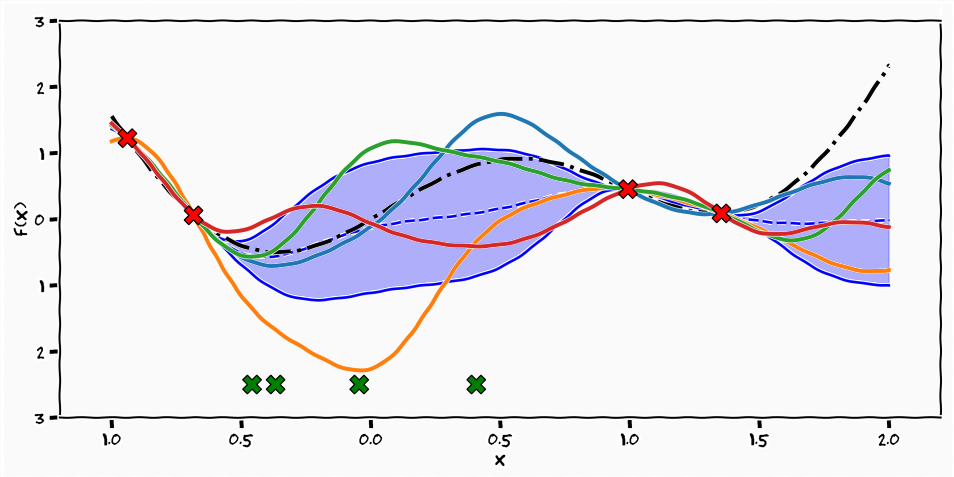

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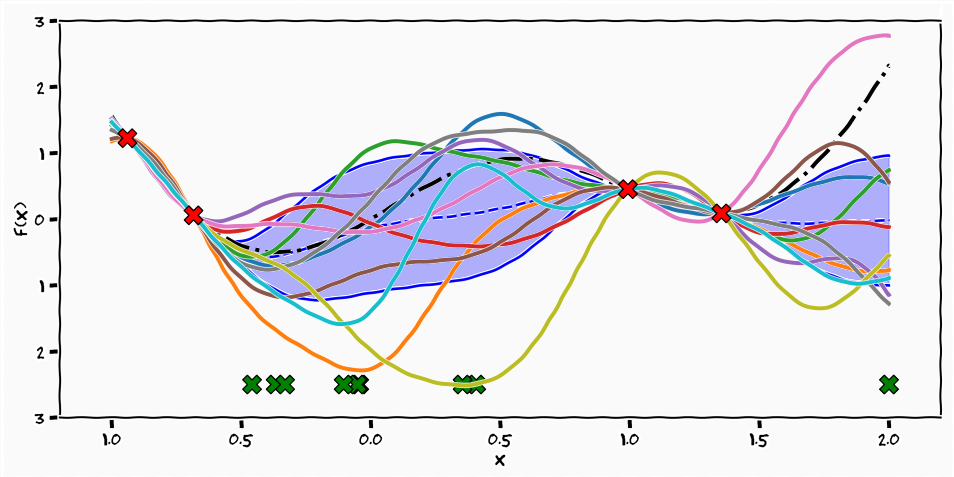

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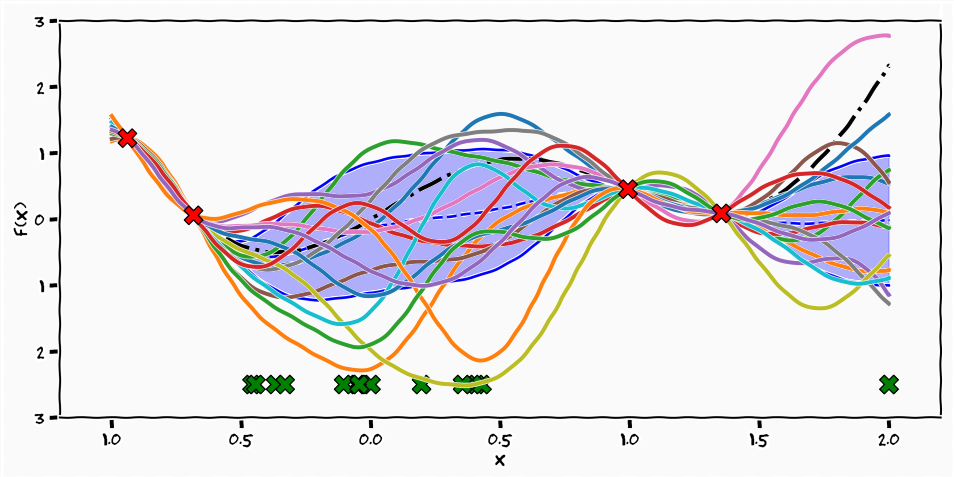
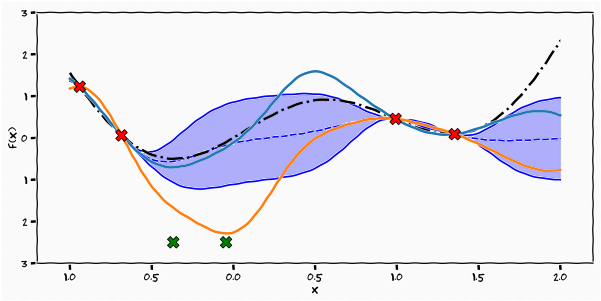


Acquisition Function

$$x_{n+1} = \underset{x \in \mathcal{X}}{\operatorname{argmax}}\, \alpha(x; \{x_i, y_i\}_{i=1}^{n}, \mathcal{M}_n)$$

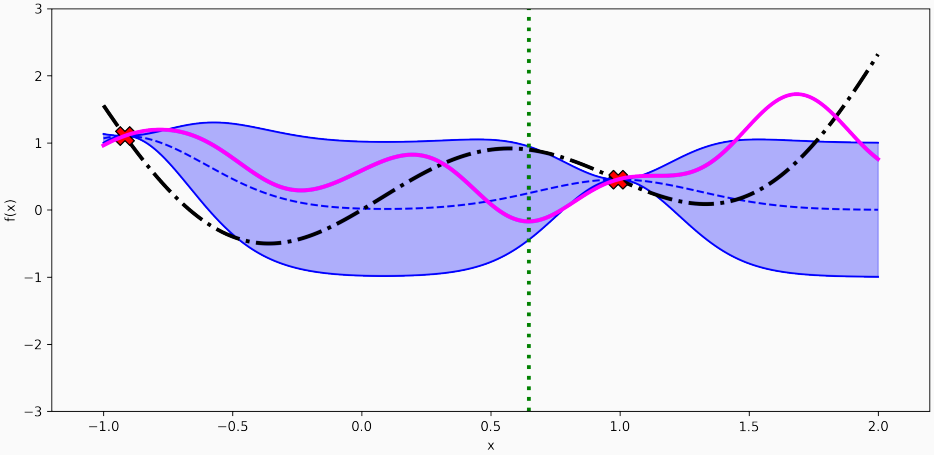
- Formulate a sequential decision problem
- This will work well if $\alpha(x)$
  - is cheap to compute
  - balances *exploration* and *exploitation*

Thompson Sampling [Thompson, 1933]



$$-\alpha(x; \{x_i, y_i\}_{i=1}^n, \mathcal{M}_n) \sim p(f \mid \{x_i, y_i\}_{i=1}^n)$$

Thompson Sampling



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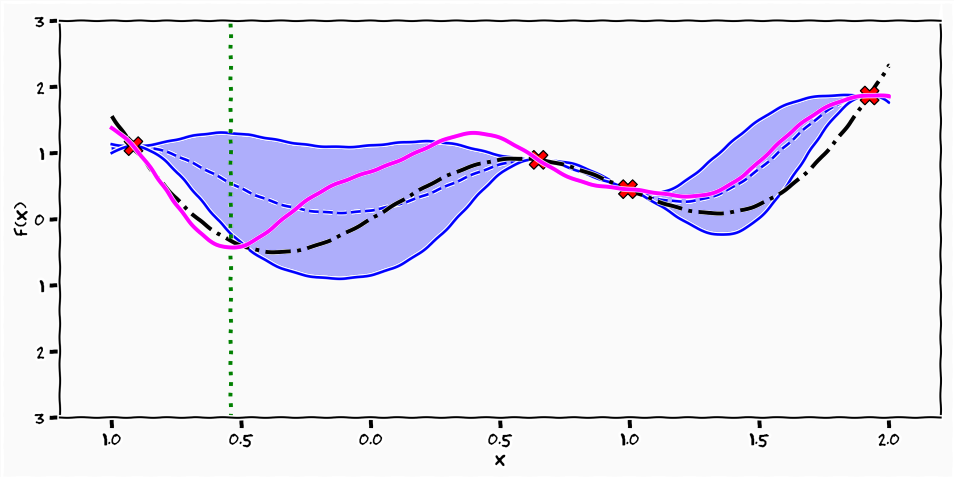

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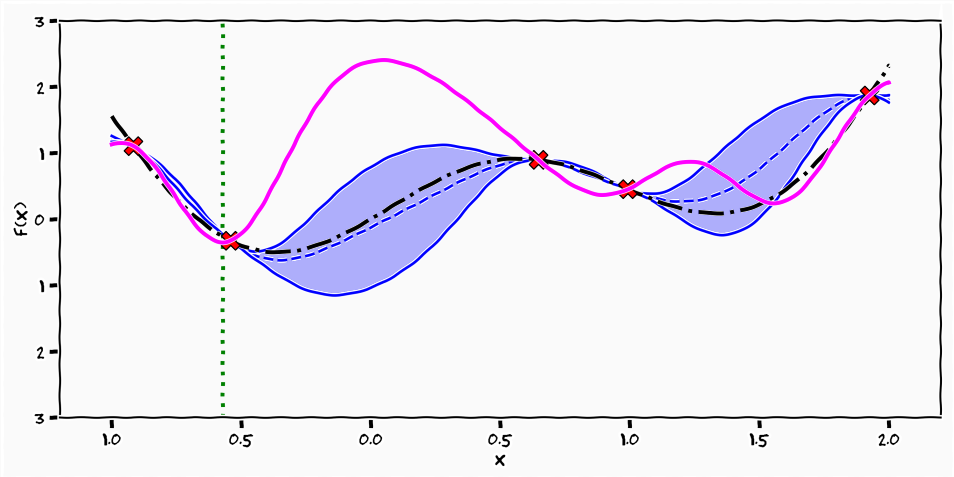

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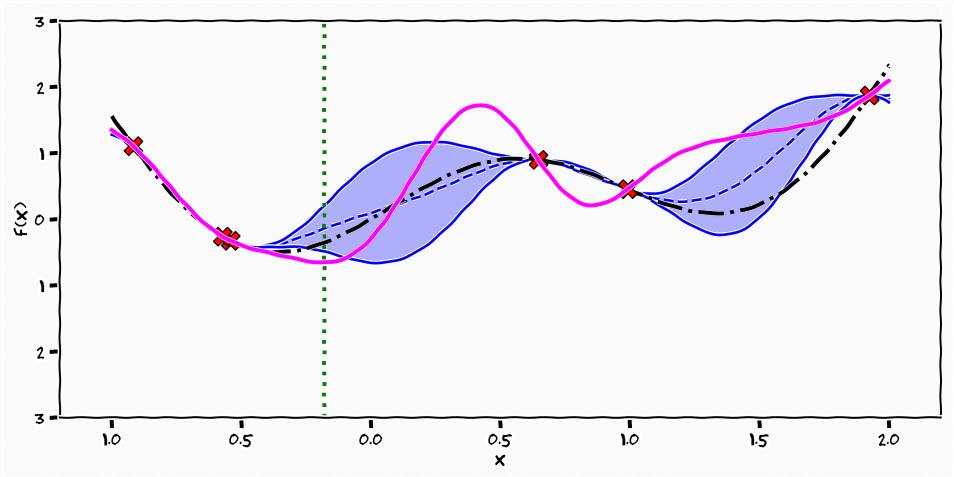

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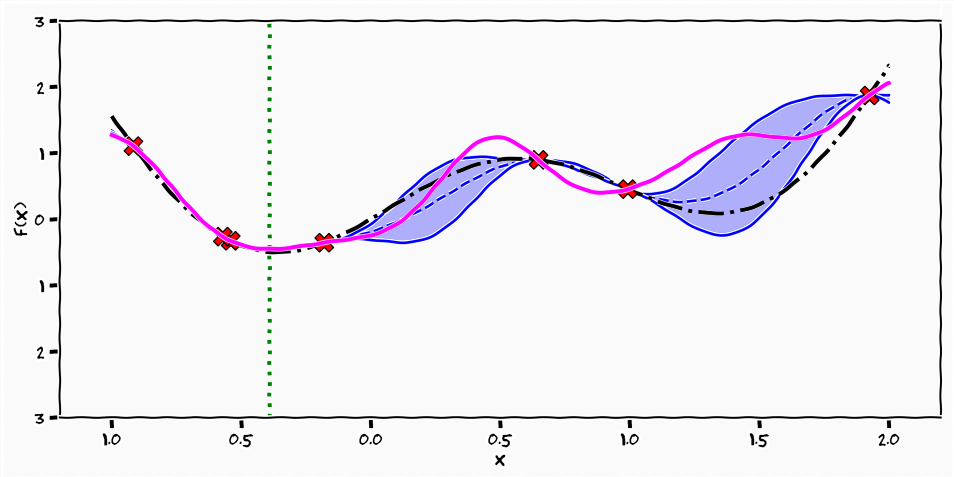

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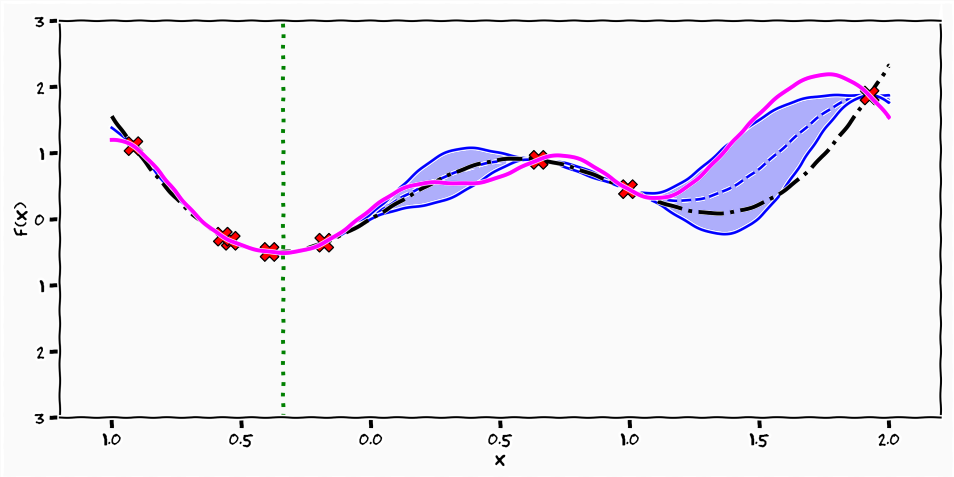

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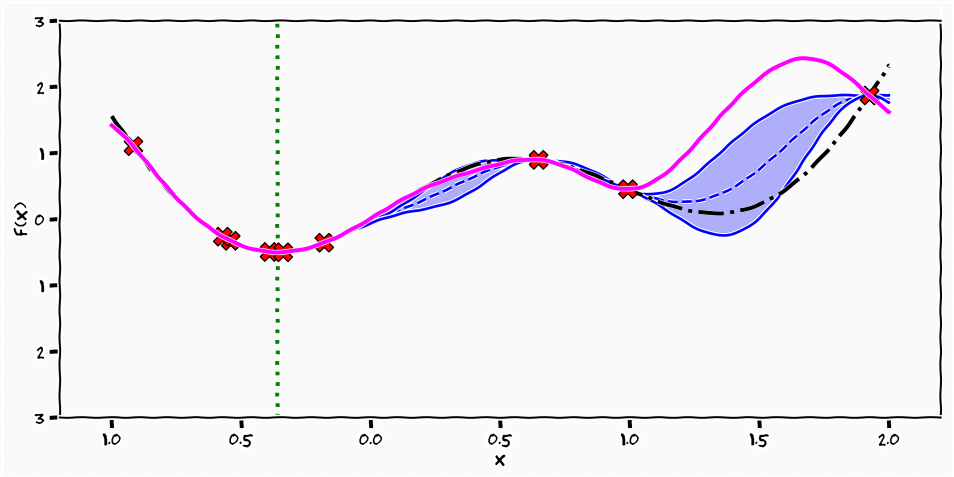
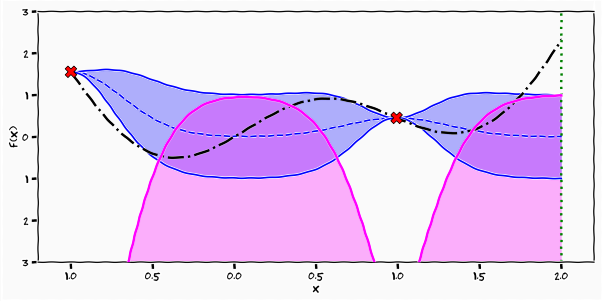


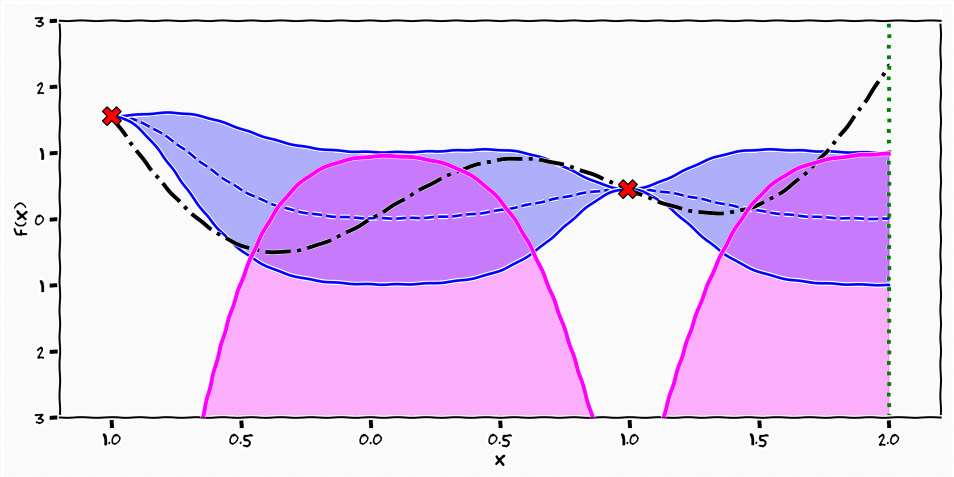
Upper Confidence Bound [Cox et al., 1997]



- Acquisition Function

$$\alpha(x; \{x_i, y_i\}_{i=1}^n, \mathcal{M}_n) = -\mu(x; \{x_i, y_i\}_{i=1}^n) + \beta\sigma(x; \{x_i, y_i\}_{i=1}^n)$$

Upper Confidence Bound



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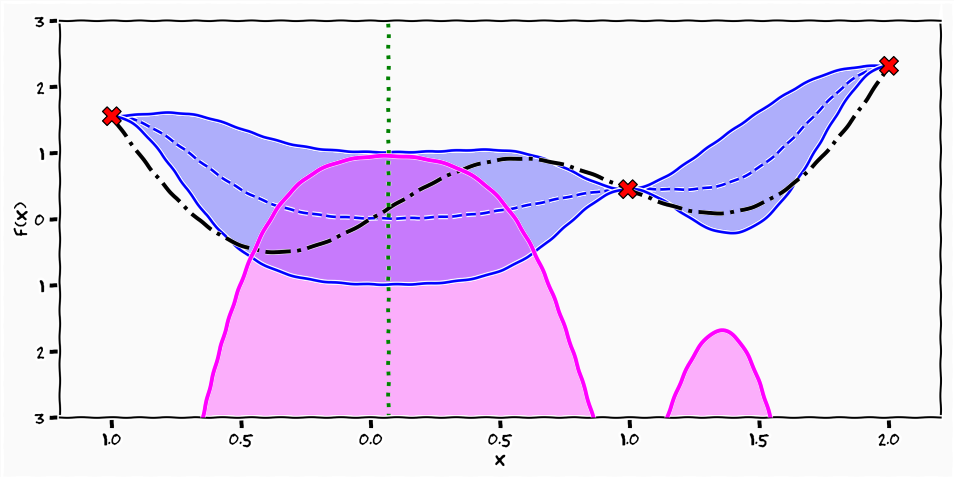

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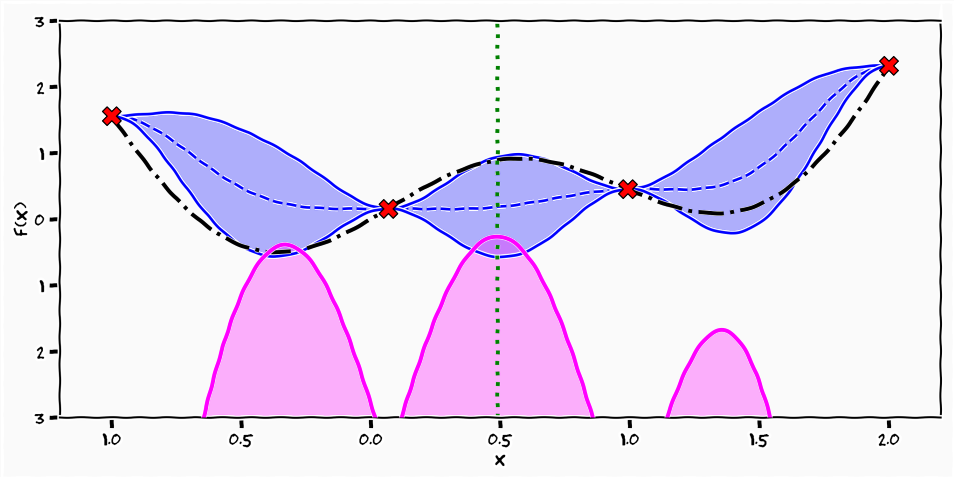

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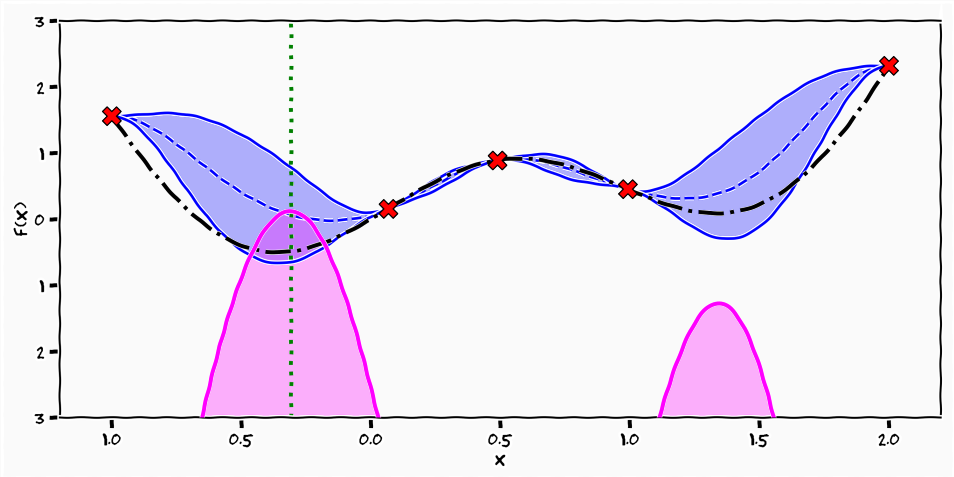
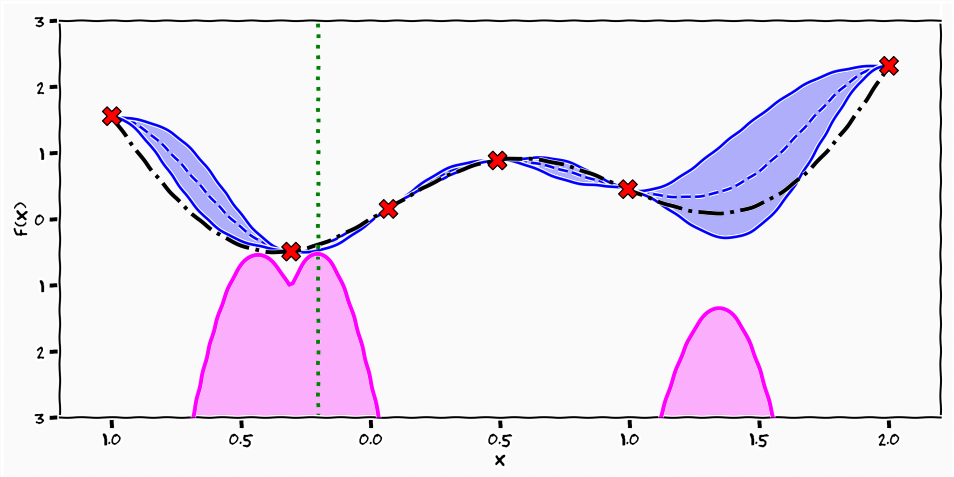
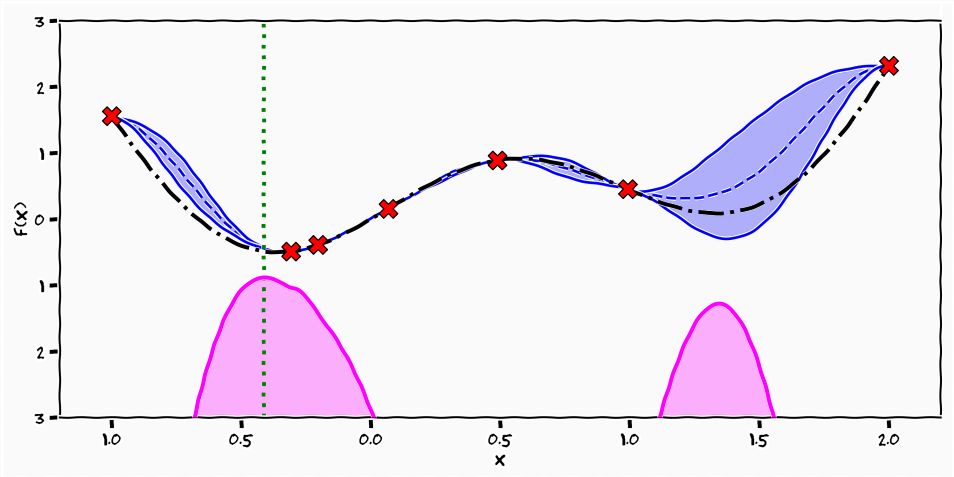


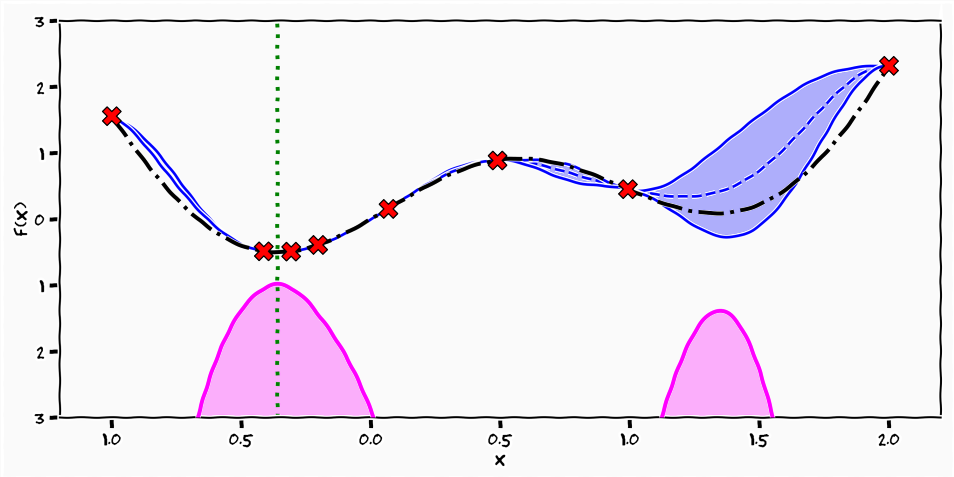
Upper Confidence Bound



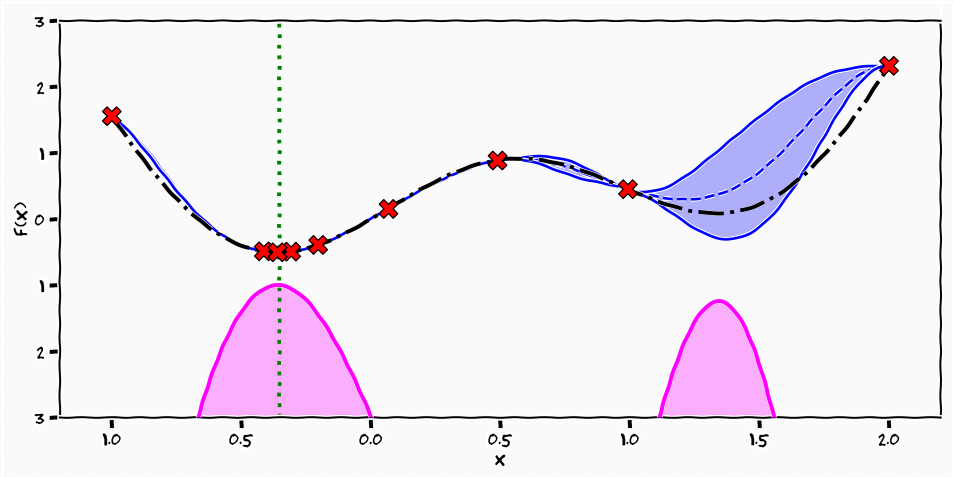
Upper Confidence Bound



Upper Confidence Bound



Upper Confidence Bound



Utility

- We can come up with lots of heuristics of how to define acquisition functions

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- Define a function that defines the utility of observing each location

$$u(x, f(x^{(*)}), \mathcal{M}_n)$$

Utility

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- Define a function that defines the utility of observing each location

$$u(x, f(x^{(*)}), \mathcal{M}_n)$$

- Define the acquisition function as the expected utility

$$\begin{aligned}\alpha(x; \{x_i, y_i\}_{i=1}^n, \mathcal{M}_n) &= \mathbb{E}_{p(f)}[u(x)] \\ &= \int u(x, f(x^{(*)}), \mathcal{M}_n) p(f \mid \{x_i, y_i\}_{i=1}^n) \mathrm{d}f\end{aligned}$$

Probability of Improvement [Kushner, 1963]

- Utility Function

$$u(x) = \begin{cases} 0 & f(x) > f(x^{(*)}) \\ 1 & f(x) \leq f(x^{(*)}) \end{cases}$$

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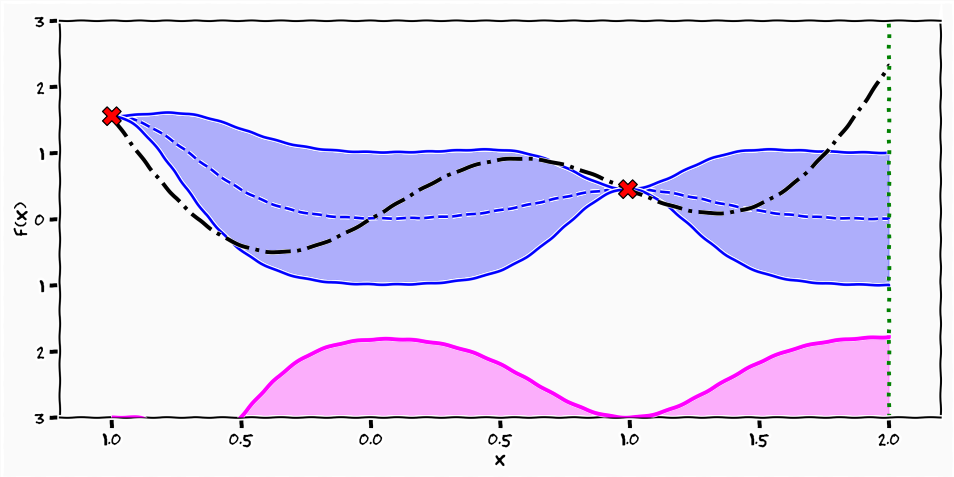
- Acquisition Function

$$\begin{aligned} \alpha(x; \{x_i, y_i\}_{i=1}^n, f(x^{(*)}), \mathcal{M}_n) &= \mathbb{E}[u(x)] = p(f(x) \leq f(x^{(*)})) \\ &= \int_{-\infty}^{f(x^{(*)})} \mathcal{N}\left(f \mid \mu(x), K(x,x)\right) \mathrm{d}f \\ &= \Phi\left(f(x^{(*)}) \mid \mu(x), K(x,x)\right) \end{aligned}$$

Probability of Improvement



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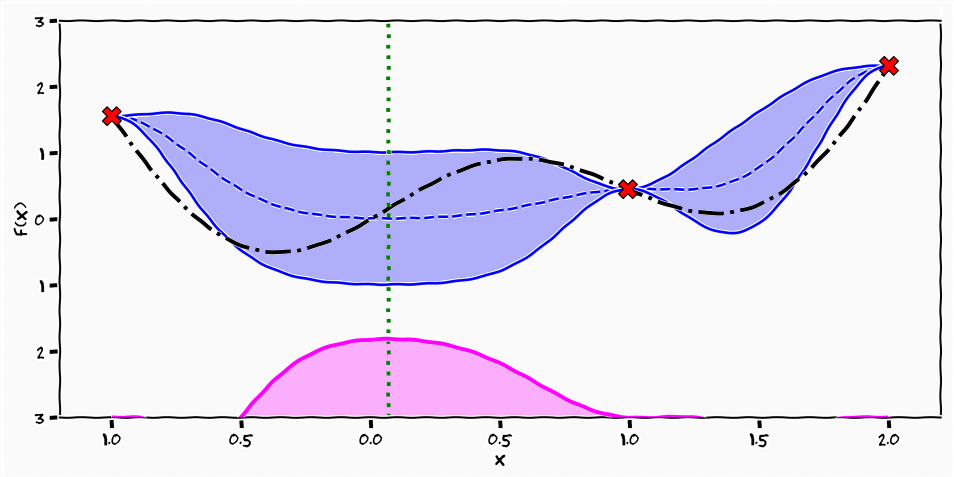

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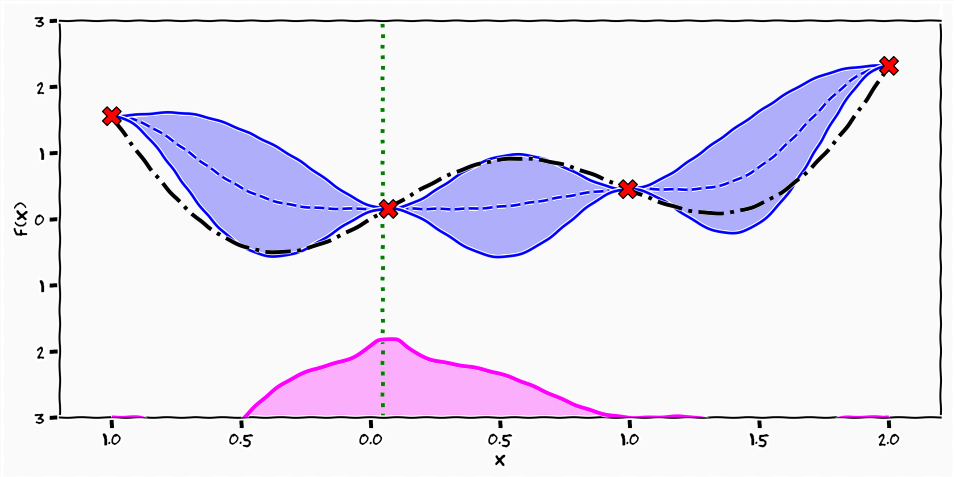

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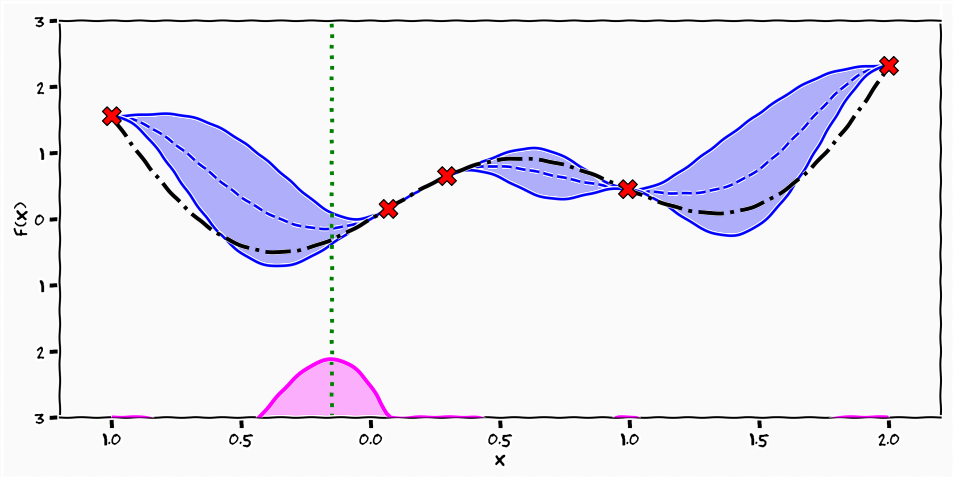

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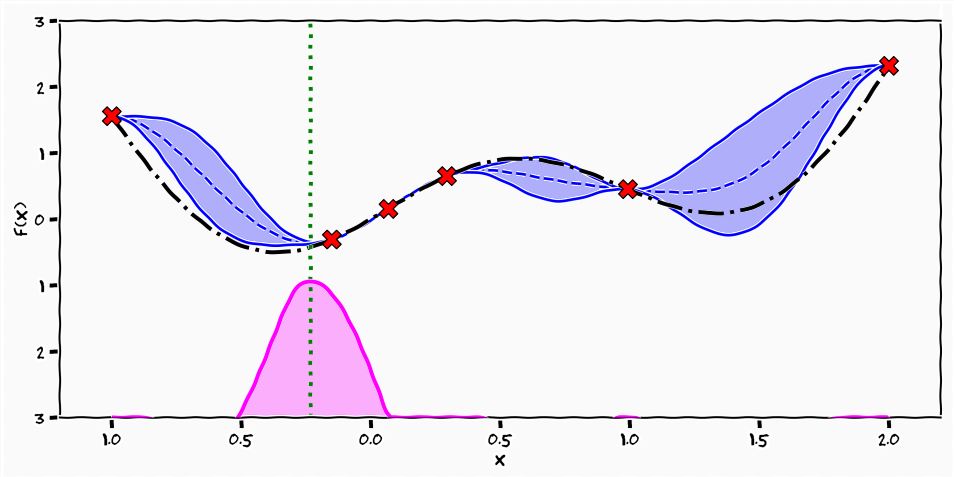

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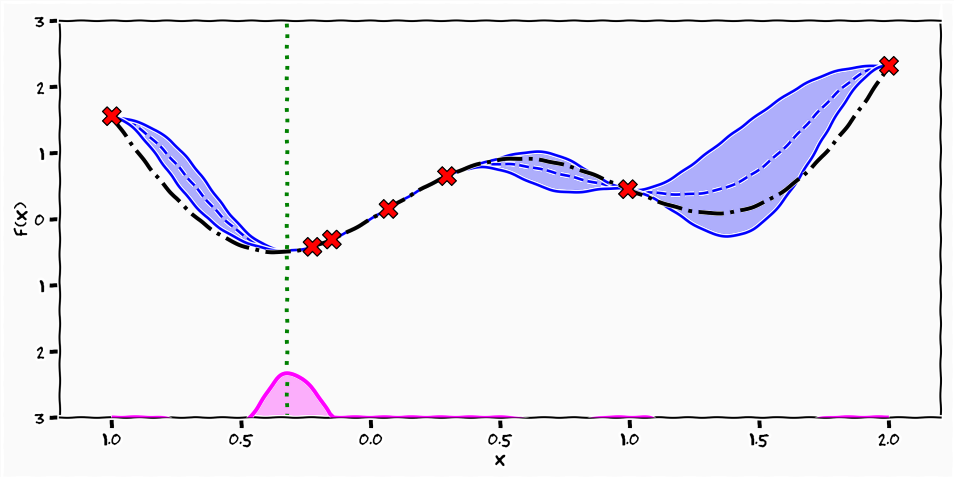

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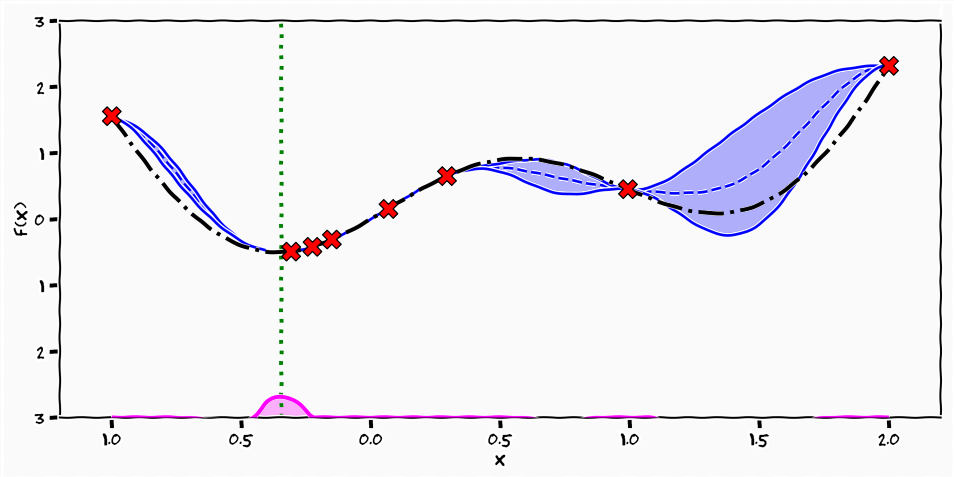
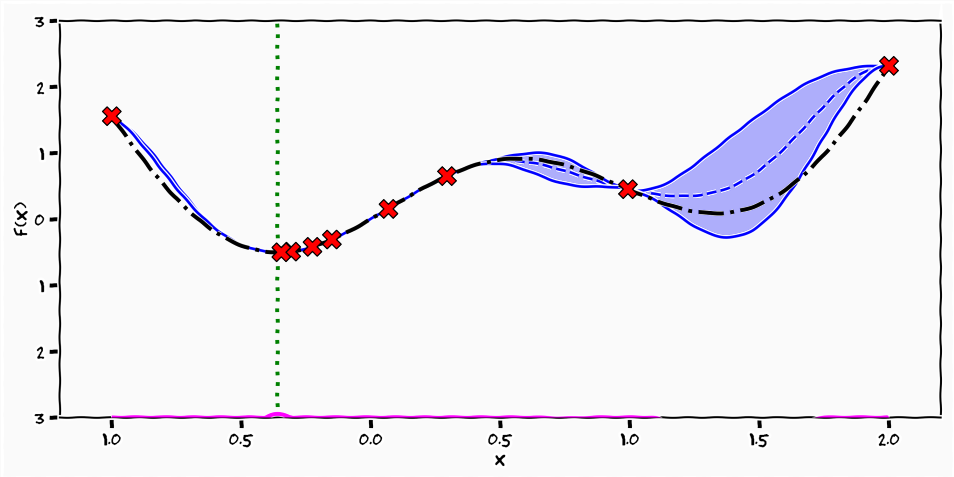


Probability of Improvement



Expected Improvement Mockus et al., 1978

- Utility Function

$$u(x) = \max(0, f(x^{(*)}) - f(x))$$

Expected Improvement Mockus et al., 1978

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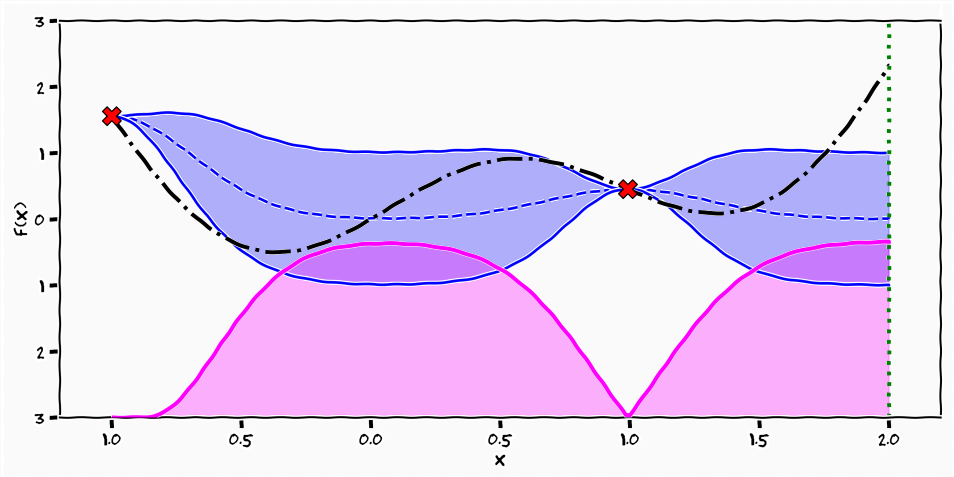
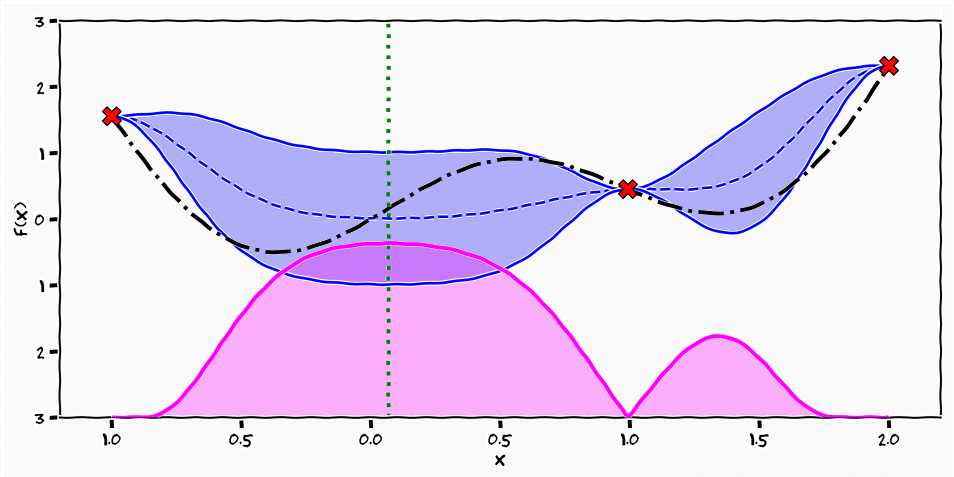
$$\begin{aligned}\alpha(x; \{x_i, y_i\}_{i=1}^n, f(x^{(*)}), \mathcal{M}_n) &= \mathbb{E}[u(x)] \\ &= \int_{-\infty}^{f(x^{(*)})} (f(x^{(*)}) - f)\mathcal{N}\left(f \mid \mu(x), K(x,x)\right) \mathrm{d}f \\ &= (f(x^{(*)}) - \mu(x))\Phi\left(f(x^{(*)}) \mid \mu(x), K(x,x)\right) \\ &+ K(x,x)\mathcal{N}\left(f(x^{(*)}) \mid \mu(x), K(x,x)\right)\end{aligned}$$

Expected Improvement



Expected Improvement



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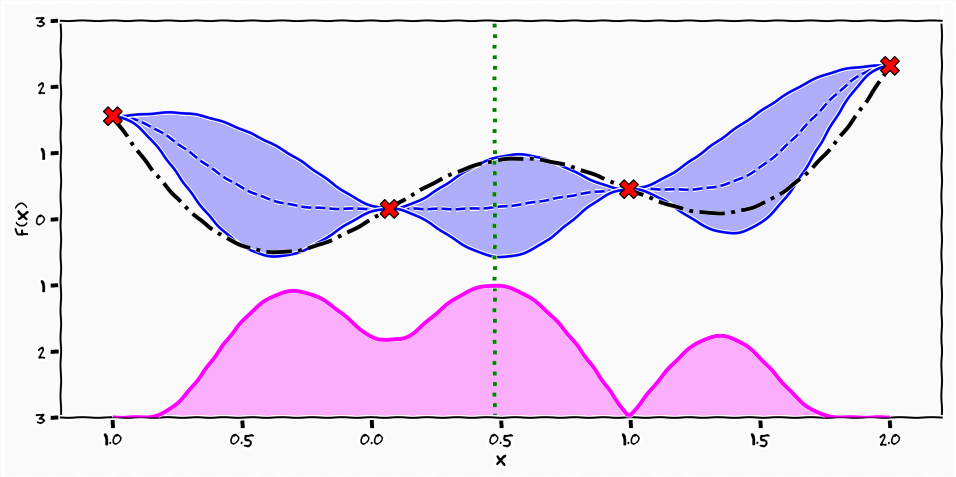

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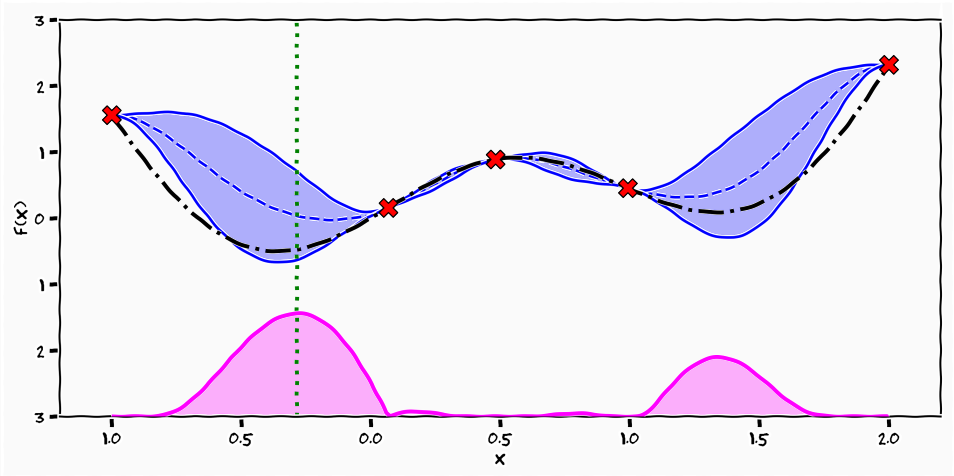

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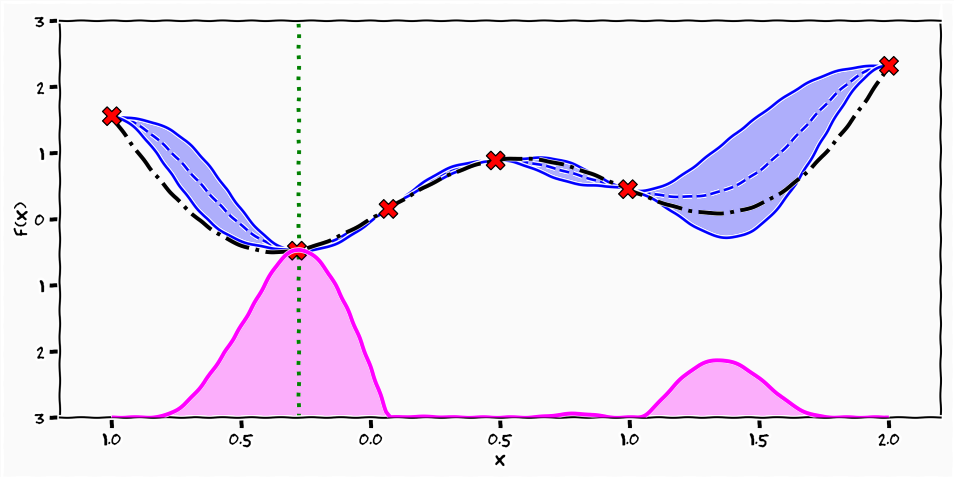
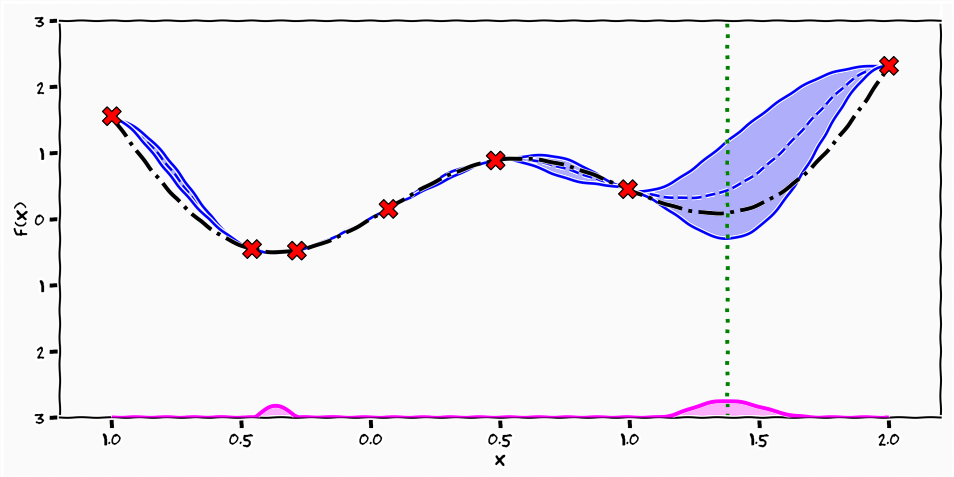
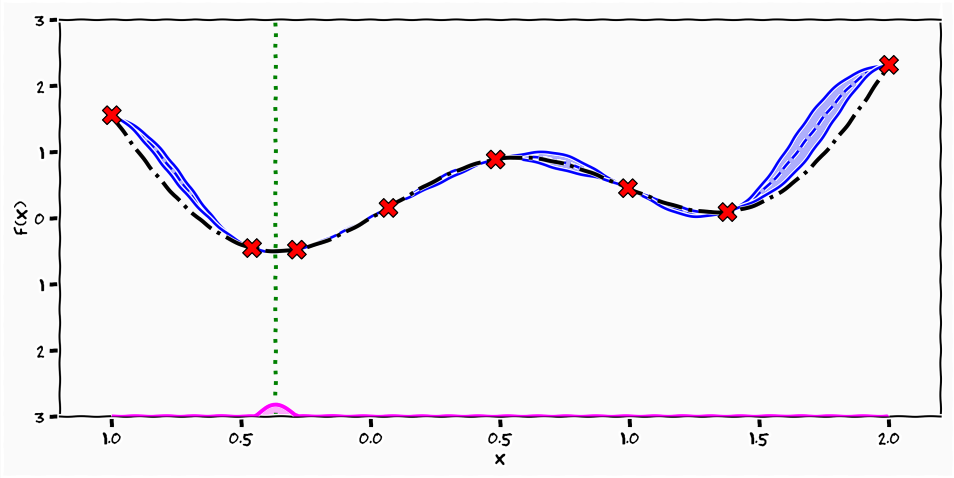


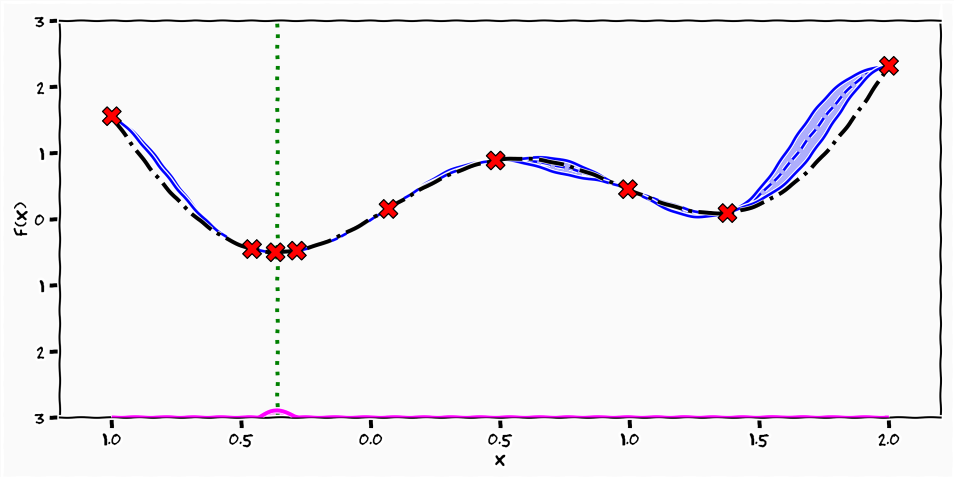
Expected Improvement



Expected Improvement



Expected Improvement



Task 1 encode your knowledge about the function in the GP prior

[1] till they open the door to the exam.

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Task 2 randomly sample some data

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Task 3 specify your acquition function

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Task 4 evaluate and maximise the acquisition function

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Task 5 add new data to model and re-estimate hyperparameters

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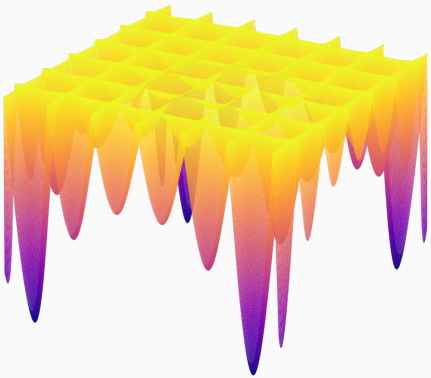
Loop 4-5 till budget is gone[1]

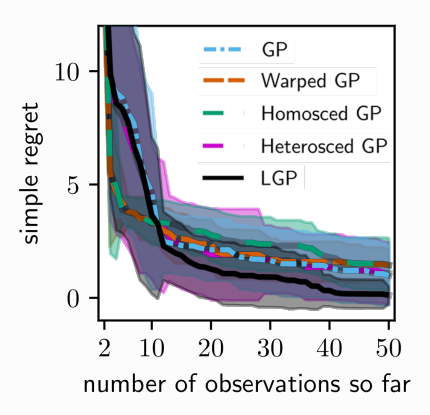
[1] till they open the door to the exam.

Bayesian Optimisation in Practice

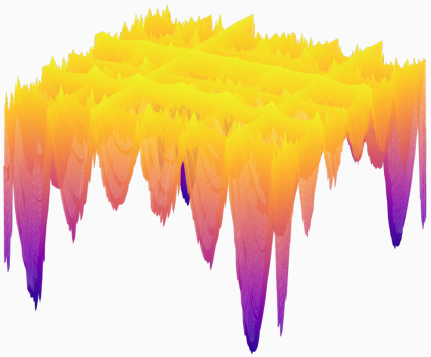
Academia vs. Industry

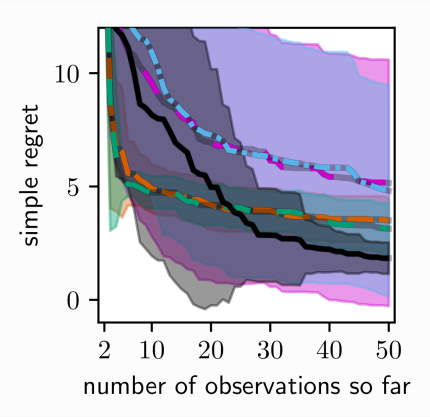
Holder Table



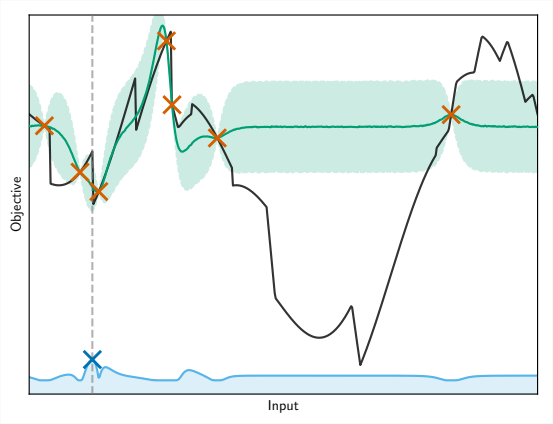


Corrupted Holder Table





Challenges: Initial Experiments [Bodin et al., 2020]



Hyper-parameters

- Fixed data

$$\hat{\theta} = \text{argmax}_\theta \, p(\mathbf{y} \mid x, \theta)$$
$$p(f_* \mid \mathbf{y}, \mathbf{x}, x_*, \theta = \hat{\theta})$$

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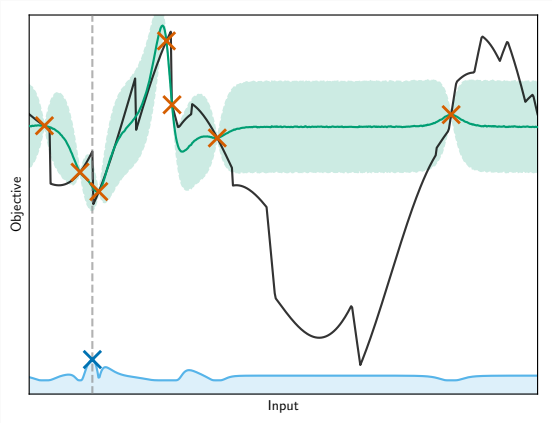
- Fixed data

$$\hat{\theta} = \text{argmax}_\theta \, p(\mathbf{y} \mid x, \theta)$$
$$p(f_* \mid \mathbf{y}, \mathbf{x}, x_*, \theta = \hat{\theta})$$

- Active setting

$$p(f_* \mid \mathbf{y}, \mathbf{x}, x_*) = \int p(f_* \mid \mathbf{y}, \mathbf{x}, x_*, \theta) p(\theta)$$

Challenges: Function is just a proxy



Challenges: High Dimensional Structures

$$k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\mathbf{x}_i^T \mathbf{x}_j}{\ell^2}\right)$$

Challenges: Greedy Acquisition

Summary

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- GPs are quite useful surrogates!

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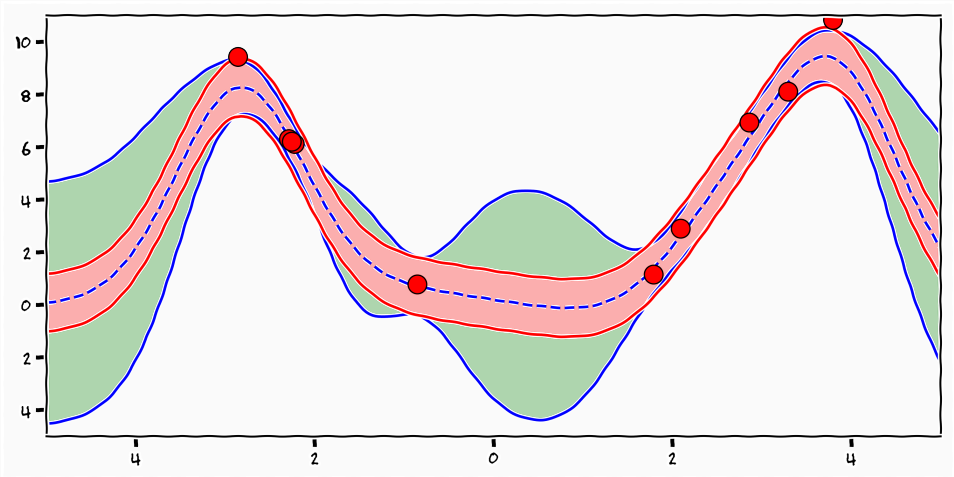
- GPs are quite useful surrogates!
- Degrees of beliefs are really useful
- The uncertainty allows us to design rich strategies for how to aquire data
- The factorisation of uncertainty allows us to describe search strategies in simple acquisition functions

”The Bayesian norm enforces hygiene between modelling and decision making.”
– Hennig, Osborne, and Kersting

Uncertainty Quantification/Factorisation

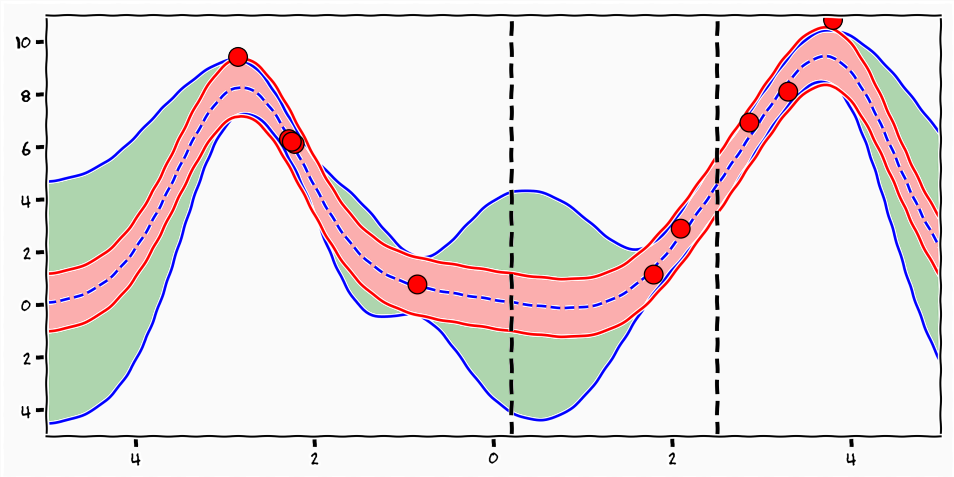


Uncertainty Quantification

Aleatoric/Stochastic ”Randomness” inherent in system, or noise in our measurement of system

Epistemic Uncertainty related to our ignorance of a the underlying system

Uncertainty for Decision Making



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